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THE REMANENT MAGNETISM OF VARVED CLAYS FROM SWEDEN

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Summary

The directions of remanent magnetization of samples taken from a number of varved clay series from Sweden have been measured in an attempt to obtain information about the direction of the Earth's field in past times. Two sets of samples were taken with a Kullenberg core sampler from cores of recent varves obtained from a river delta. It was possible therefore to compare the magnetization directions of these samples with observatory records of the Earth's field direction at the time of deposition of the clay. The results indicate that the magnetizations were acquired on deposition and have remained stable in direction since. Their directions are shown to have been controlled not only by the Earth's field but by the slope of the bed on which deposition took place and very probably by bottom currents flowing in the river or lake during sedimentation. Thus no information about the past direction of the Earth's field can be obtained from the data unless corrections can be applied to allow for these factors. A method of correcting approximately for the change of magnetization direction due to deposition on a slope is explained. No way of allowing for the effects of bottom currents has yet been discovered, though there is evidence that in some instances these effects are small. Where this is so the magnetization direction may perhaps prove a reliable indicator of the field direction at the time of deposition.

Introduction.—If the remanent magnetism of sedimentary rocks is to yield information about the geomagnetic field in past times then the rocks must satisfy certain requirements. The period of their history in which the magnetism was acquired must be known, and its direction must still be the same as the geomagnetic field at that time or be related to it in some ascertainable manner. The investigations to be described were an attempt to find out whether varved clays satisfy these requirements.

For such a purpose varved clays possess a number of advantages over older and more consolidated rocks. They show an annual banding by means of which they have been accurately dated; time intervals between layers can thus be determined, and exact correlation of deposits at different localities made. They have had a simple geological history, and being also comparatively recent sediments, their initial polarization has had the best chance of survival. Finally, as they are unconsolidated, they can be redispersed and resettled under laboratory conditions where their behaviour during sedimentation can be closely studied.

Some work on varved clays had been done previously. In 1938 McNish and Johnson (1) reported the results of measurements of the remanent magnetization of some New England Clays. This was followed in 1947 (2) by a further study of varves from different localities covering a period of 5000 consecutive years, together with laboratory studies on artificially deposited clays. In their opinion the results indicated that the horizontal polarizations of the varves represented the direction of the changing Earth's field at the time of deposition of the clay but that the inclinations of the total polarization were unreliable. They considered the magnetization to be due to the alignment on settling of contained magnetic particles. Previous to these experiments Ising (3) had published an account of measurements on a 350 year long section of Swedish varves. The directions of the horizontal components of the polarizations agreed with the present field direction to within a few degrees, the inclinations being considerably less than that of the present field. Ising considered that the measurements indicated that the clays were unstable.

The results of these various investigations were therefore contradictory, and no certain conclusion could be drawn from them as to how closely the polarization directions of sediments reflect the direction of the magnetic field at the time of their deposition. A direct way of obtaining such evidence appeared to be a comparison of polarization directions of varves laid down in the last few hundred years with observatory records of the field for the same period. In Scandinavia such varves can be obtained only from the deltas of rivers draining glaciated areas, and have to be collected with core samplers. In 1952 such cores were collected with a non-magnetic Kullenberg core sampler carrying an orientation device.

Another but less direct method of testing the reliability of polarization measurements seemed to be the collection of contemporaneous sequences of older varves from exposures on land at different localities in order to find out if any correlation existed between their respective polarization directions. Several collections for this purpose were made in Sweden between 1950 and 1952.

Collection of the varved clays.— To obtain an accurate value for the direction of polarization of a clay layer two separate observations have to be made. The orientation of reference axes marked on the sample must be measured with respect to the Earth's field before the sample is removed from the body of the clay, and then the relation between the polarization direction and the reference axes has to be determined. The sampling technique must not distort the clays, and their magnetization must not be altered during transport to the laboratory.

The sites selected in Sweden were steep river banks where thick varve sequences had been exposed by erosion. The procedure was to dig in the bank a series of steps having vertical back walls. The fresh clay exposed in these faces was then scraped smooth and made exactly vertical with brass tools, and the varve layers identified and counted.

On the first visit to Sweden the clay was collected in 2 ft. long duralumin channels, these being eased into the clay in a vertical position, and the orientation taken with a prismatic compass and clinometer. Later, in the laboratory, cube-shaped specimens were cut from the troughs and transferred without loss of orientation to small perspex boxes of $\frac{7}{8}$-inch internal diameter. For this purpose a special device was built consisting of a plate to which the channel can be rigidly attached, this being straddled at right angles by a bridge. The bridge

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carries a vertical square-section brass tube which is pressed into the clay to obtain a sample. A piston inside the tube permits the specimen thus collected to be ejected into a perspex box. A similar device was used to collect specimens from the tubes of the Kullenberg core sampler, the tubes being first opened by cutting away about a third of the circumference.

The device was also adapted to the direct collection of samples in the field, either from a vertical clay wall or from a horizontal clay strata plane. In this modified apparatus the sampling tube slides in a slot in a frame which is fixed to the clay by four long pins. The frame carries a prismatic compass and two levels so that its orientation can be determined. The orientation is estimated to be correct to better than $\pm 2°$.

When collecting, careful measurements of the geological dips of the layers were made. The dips are estimated to be accurate in amount to about 2°, and the strikes to about 10°.

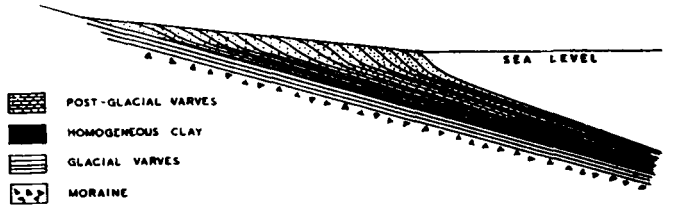


FIG. 1.—*Section of varved sediments along a river valley.*

For the purpose of the magnetic investigation the varve series to be collected had to fulfil a number of conditions. Their age had to be known, and there had to be available already correlated contemporaneous sequences from localities at least a few kilometres apart to ensure that the conditions of deposition of the two series had been somewhat different. It was also very desirable that at each locality the sequence should be relatively undisturbed and contain preferably several hundred years of varves. The post-Glacial varves of northern Sweden, now exposed in certain river valleys, satisfy these conditions. These sediments were laid down after the ice of the Pleistocene glaciation had retreated to the mountains. At that period the Baltic stood at a higher level than it does today, and many of the northern river valleys were fjords. Rivers draining the remnant of the ice-sheet emptied their loads into these fjords, building large deltas. Because of the annual variation in stream load, due to the sub-arctic climate, the delta deposits are annually banded, and are called varves. Near the river outlet the varves are thick and sandy, but they thin outward from the delta and pass into fine silt and clay. As a consequence of the isostatic uplift of the land these deltas slowly travelled downstream. Thus the post-Glacial succession at any locality begins at the bottom with homogeneous clay, this passing up into thin clay and silt varves. Still higher these become more sandy, the succession ending with a metre or so of sand. Deposition of varves is, of course, still proceeding in the present river deltas. Below the homogeneous clay is found a series of much older glacial varves, laid down when the ice-front itself retreated past the area. A diagrammatic profile of a typical succession is shown in Fig. 1.

The rivers have now eroded through the deposits, and good sections are available in their steep banks. Although the age of the post-Glacial varves

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decreases from the upper to the lower part of the valleys, sections a few kilometres apart can be found which have several hundred years of varves in common. Unfortunately the older Glacial varve sequences are usually much shorter.

To commence the investigation a thousand year varve series was collected at Prästmon on the Ångerman River. The varves throughout this valley had been dated by Lidén (4), and by reference to old maps and records he had been able to show that the topmost varves here are about 1000 years old. A similar thousand year series was then collected at Undrom, a locality a few kilometres upstream from Prästmon. This series was older, but overlapped the Prästmon varves by about 400 years. Owing to the difficulty of distinguishing boundaries in the thin varves only an approximate correlation between the two localities was obtained. The estimated maximum error is thought to be not more than fifty years.

A year later varves of Glacial age were collected from sections in the vicinity of Ragunda on the Indal River. A number of factors governed this choice. It was thought that these varves may have been deposited under quieter conditions than the post-Glacial varves. They had also the advantage of being easily accessible in a number of undisturbed sections for which the chronology had already been worked out. Material was collected at Lien and Bispfors, two localities about ten kilometres distant from each other. Two sections a few hundred yards apart were cut and sampled at each locality. The series were almost exactly contemporaneous at all four sites, but the period covered was only a hundred years.

In the same year continuous varved clay cores were obtained from the present delta of the Ångerman River near Kramfors, covering approximately the period A.D. 1300–1900. Two cores were made available to us at Birmingham, the orientation of only one being known owing to a failure of the orientation device. The varves were dated by comparison with varve diagrams of other cores previously sampled at the same locality and dated by Dr From of the Swedish Geological Survey (unpublished work).

Measuring apparatus.—The clays to be measured were known to have magnetic moments of between 10^{-4} and 10^{-6} c.g.s. units per c.c. Using samples of about 1 c.c. and measuring to an accuracy of 2° this meant building an instrument capable of detecting moments at least 25 times less, i.e. of 4×10^{-7} c.g.s. units. The a.c. magnetometer designed by Johnson (5) was adopted as a basis for the Birmingham instrument, though in its final form it is somewhat different. The circuit diagram is shown in Fig. 2. In essence it consists of a coil, very near to which the specimen is rotated at 30 revolutions per second. The specimen can be considered to approximate to a magnetic dipole, and as the plane of the coil is parallel to the rotation axis, the e.m.f. induced by the dipole is proportional to the component of the specimen moment in a plane at right angles to this axis. From measurements of the phase and magnitude of this e.m.f. the intensity and direction of the dipole component of the specimen can be obtained provided the instrument is first calibrated.

With moments of the order of 10^{-5} units the induced e.m.f. is of the order of a microvolt, and refined methods have to be adopted to measure it. The signal is amplified and then fed to a synchronous rectifier. This consists of a cam-operated switch on the end of the shaft. This switch can be turned about the rotation axis, its position being read on a circular scale of degrees.

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The switch earths the amplifier output for alternate half-cycles, the portion of the cycle chosen depending on the switch position. The output is connected by way of a resistance capacity filter to the grid of a twin triode, the second grid of which is connected to a potentiometer across the high-tension supply. Measurement of the potential applied to the first grid is made in terms of the current flowing through a microammeter connected across the two cathodes. The sensitivity of the apparatus depends on reducing the band-width of the system, this being done by increasing the time constant of the filter. The output of the synchronous switch is a sine function of its position, and so the magnitude and phase of the coil output can be found by setting the switch to give a zero reading of the microammeter, and then turning through 90° to get the amplitude.

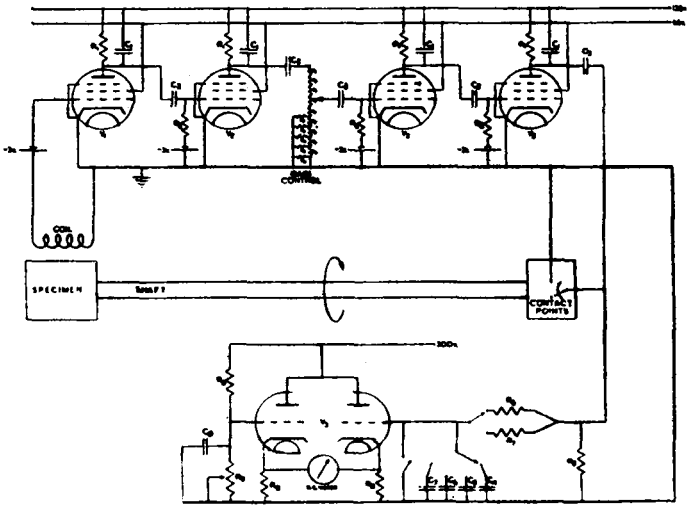


FIG. 2.—*Circuit details of magnetometer.*

A number of refinements had to be made to improve the sensitivity. In order to minimize the effect of uniform but fluctuating external stray fields an antiphase winding, having the same area turns as the inner coil, was wound outside it, and connected in opposition. The e.m.f.'s induced in the two coils by such fields thus tend to balance out, their disturbing effect being reduced to a minimum by adjustment of the antiphase winding. Because of its larger diameter the e.m.f. induced in this coil by the specimen is not large, the net effect being an improvement in the signal-to-noise ratio. The coils are also shielded from external fields by enclosure in a thick aluminium box. It was found necessary to use a non-metallic shaft and specimen holder, all the metallic materials tried generating small e.m.f.'s in the detector coil, either because of ferromagnetic impurities, or because they were conductors. Non-conducting materials, however, produced electrostatic charges, these also generating spurious and fluctuating signals. These undesirable effects were very much reduced by coating the specimen holder and the coil with a thin conducting film. Certain other sources of noise were eliminated by careful screening and earthing.

A high gain, low frequency, resistance-capacity coupled amplifier is used, this being battery operated throughout.

With a time constant of 12 seconds the minimum detectable moment is about 5×10^{-7} c.g.s. units.

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Generally speaking the apparatus is satisfactory, though it has a number of faults. The most serious is the mechanical difficulty of keeping the contacts of the synchronous switch so adjusted that whatever its position it opens for exactly 180°. This setting is most delicate. Failure of adjustment leads to errors of up to a few degrees in the measurement of directions. Another difficulty is that of reducing the spurious e.m.f.'s generated by the spinning specimen holder which have already been mentioned. Normally the magnitude of these is perhaps only a few per cent of the specimen signal, but they set a limit to the accuracy with which weak specimens can be measured.

Results.—A short summary of the Ångerman River varve measurements has already been published (**6**). They are given again here and discussed more fully as later work has made their meaning clearer. About 150 samples were measured, roughly equally divided between the two localities of Prästmon and Undrom. Much of the material was sampled at about 20 year intervals, but there was some variation from this. The total interval covered by the samples was from about 1100 B.C.–A.D. 750, the period covered by the overlap being from 150 B.C.–A.D. 250. The directions of the total polarizations, plotted on a stereographic projection, are shown in Fig. 3. (Lidén's time scale is used on

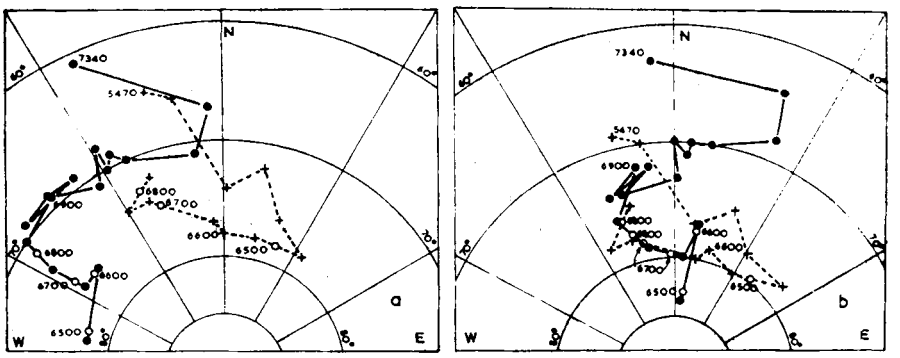


FIG. 3.—*Stereographic projection showing polarization directions of varves from Prästmon and Undrom. (a) Uncorrected and (b) corrected for dip. Dates in years on Lidén's Time Scale on which 7602 equals A.D. 1000.*

Prästmon — ● — *Undrom* – + – *Interpolated points* ○.

the diagram.) The scatter of individual measurements is rather large, and to make relationships clearer the points plotted are the average of measurements on five consecutive samples. Each point thus represents a mean value of the polarization direction for an interval of up to 100 years. The radii of the circles of confidence for the 95 per cent level of significance, calculated by a method due to R. A. Fisher (**7**), are about 4°. It will be seen that there is a considerable difference between the polarization directions of varves of the same age at the two localities. However, if the assumption is made that the varves were horizontal when deposited, and the magnetic vectors rotated to correct for the geological dips of the beds (almost 3° at Undrom, and varying from 11° to 13° at Prästmon), then quite good agreement is obtained between the two sets of values for points of corresponding age. Statistical treatment of the data is hardly justified, but a rough measure of the improvement in agreement obtained on correction, and of the closeness of fit, can be made by comparing differences. Points have been interpolated representing polarization directions for varves

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of average age 6500, and at successive hundred year intervals to 6800, and the mean difference between these interpolated points measured, both before and after correction. The process of correction reduces this mean difference from 13° to 3°. The individual differences are thus reduced to values somewhat less than the corresponding circles of confidence, and thus it seems unlikely that the correspondence is fortuitous.

Generally speaking the range of the polarization directions after correction, and their rate of change, are compatible with the view that they were controlled by the geomagnetic field during deposition of the clay. Moreover the agreement implies a fair degree of stability of the magnetization directions. It also necessitates magnetization during, or shortly after, deposition. There are, however, some features of the results which make qualification of these statements necessary. For example, in the youngest Prästmon varves, the inclinations seem too low, and the rate of change of direction abnormally high, for secular variation to be the only controlling factor. It seems probable therefore that during deposition of some parts of the varve sequence the sedimenting magnetic particles were affected by other forces in addition to the geomagnetic field. The effect may of course be a post-depositional one, though evidence which is discussed later points to the former cause.

A further corollary of the agreement of polarizations at Undrom and Prästmon appears to be that both sets of varves were laid down horizontally, for this agreement only appears after correction of the varves back to a horizontal plane. The field evidence, on the contrary, indicates that the beds were part of a delta, the dip being depositional. This contradiction has been resolved by sedimentation experiments, and the dip correction as a first approximation is now known to be justified.

Comparison of polarization direction of the Ångerman River delta cores was in every way a comparable experiment to the above. The cores appeared similar in character to the Prästmon and Undrom varves, consisting of thin silt and clay varves at the lower end, these giving place to thicker and more sandy varves near the top. There was evidence of disturbed conditions at times during sedimentation, and the varves in one core (Core III) had variable dips, though these were not greater than about 5°. The varves in the other core (Core I) were almost horizontal. There was some distortion of the cores round the circumference due to the sampling process, but it was always possible to extract from their centres small apparently undistorted samples. In all about 130 samples were taken from the two cores. The cores were in half-metre sections, and between 6 and 8 samples were taken from each length at equal spatial intervals. Average directions of polarization were plotted for the sets of samples from each core length, time intervals between these values increasing from about 35 years near the top of the core to about 100 years at the bottom.

Fig. 4 is a stereographic projection of the corrected total polarization directions for Core I and Core III. The orientation of Core III has been obtained by assuming that it was sampled vertically, and that the average declination of all the Core I and Core III measurements for the same time interval is the same. The radii of the circles of confidence for the 95 per cent level of significance lie between 7° and 10°. Since the range is about the order of magnitude of the expected secular change, little information about this can be expected from the measurements. A general trend from east to west, with decreasing inclination, is suggested, and in general the polarization directions are not inconsistent with

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the field directions for the corresponding dates, excepting perhaps in the youngest varves of Core I.

Fig. 5 (*a*) shows the combined data for both cores expressed as declinations and inclinations, together with the known secular variation of the declination. The best straight line has been plotted through the declination measurements for the period from A.D. 1550 to A.D. 1800, measurements on younger varves being excluded since these give less consistent results than the earlier clays. This line parallels the known declination change, though owing to the scatter it is of doubtful significance. Fig. 5 (*b*) shows the corresponding inclination changes.

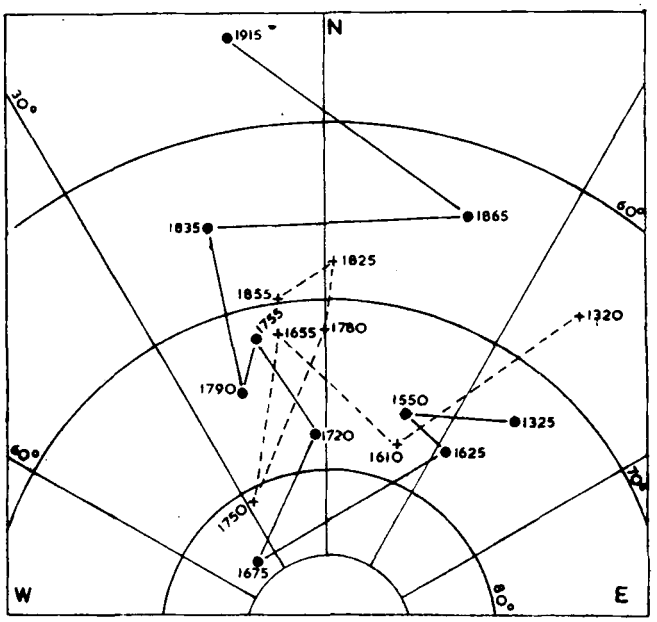


FIG. 4.—*Stereographic projection showing polarization directions of varves from the Ångerman River cores corrected for dip. Dates in years A.D.*
Core I – – + – – *Core III* — ● —

From the point of view of confirming whether or not the polarizations of varved clays reflect the changing direction of the Earth's field the measurements are unsatisfactory. If the scatter of the polarization directions were random about the true field direction, then it might be possible to obtain a reasonable idea of the field change by measuring a very large number of samples. There are, however, reasons for believing that there are systematic deviations of the polarizations from the true field direction. For example there seems to be somewhat better agreement between corresponding polarization directions for the two cores than between either set and the corresponding direction of the geomagnetic field, though it is not possible to be sure of this because of the large uncertainty in the positions of the plotted points. The circles of confidence are considerably larger for the polarization measurements on the recent sediments than they are for Prästmon and Undrom. Since all the sediments come from the same river it is reasonable to infer that the cause of the scatter is the same in all instances. As the scatter is greater in the more recent varves, and particularly in the sandy variety, the effect is more likely to have a hydrodynamic than a

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magnetic cause. It seems likely therefore that hydrodynamic forces, most probably due to currents acting during sedimentation, are responsible for systematic deflections of the polarization from the Earth's field direction. The cores were collected at points quite near each other, at which the velocity and direction of any currents off the delta might be expected to be substantially the same at any particular time. Consequently any changes in polarization direction brought about by changes in such currents would be more or less

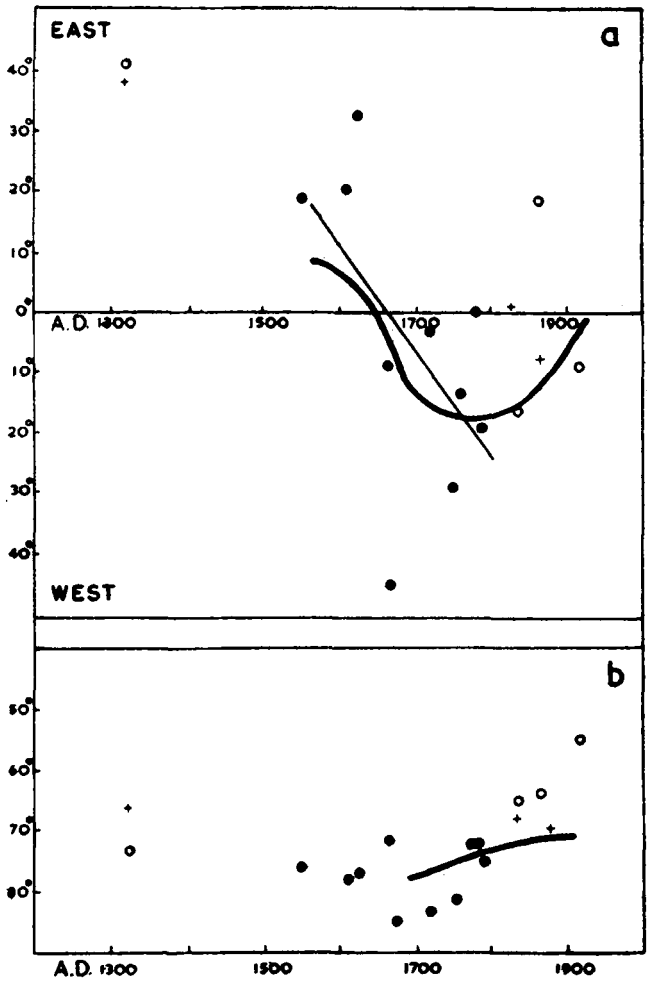


FIG. 5.—(*a*) *Declination and* (*b*) *Inclination of polarization directions of varves from the Ångerman River cores.*

Points used in fitting best line shown thus : *Core I* ⊕, *Core III* ●.
Other points shown thus : *Core I* +, *Core III* ○.

the same at both localities. Some local variations would almost certainly occur, and could account for the lack of a good correlation. (Measurements at present being made on some recent Icelandic varves strongly suggest that this explanation is correct. Here the corresponding declinations beween series of varves taken from two localities about one hundred yards apart are very close. The declinations themselves are quite unrelated to the known field change.) On the other hand a scatter which was due to post-depositional reorientation of magnetic

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particles, or to the decay of the magnetization of such preferentially orientated particles, would probably increase rather than decrease with the age of the deposit.

Some preliminary experiments indicated a high magnetic stability for the clays. Re-measurement of some Prästmon samples which had been lying with random orientations in the Earth's field for over a year revealed no significant change in direction. The coercivity of the natural remanence is very high, a.c. fields of 1000 gauss being insufficient to destroy it. In this connection it may be mentioned that the varves appear to contain both a " hard " and a " soft " magnetic component. They are easily magnetized in a field of, for example, 100 gauss but this magnetization can be destroyed by an a.c. field of the same strength, leaving the original remanence unaltered. Separation of the magnetic components of the varve material shows that the magnetic particles vary in size from about 20 μ downwards, the lower limit probably being below the resolving power of the microscope at about 1 μ. Many of the particles appear as inclusions in quartz and other minerals. The material has not yet been identified.

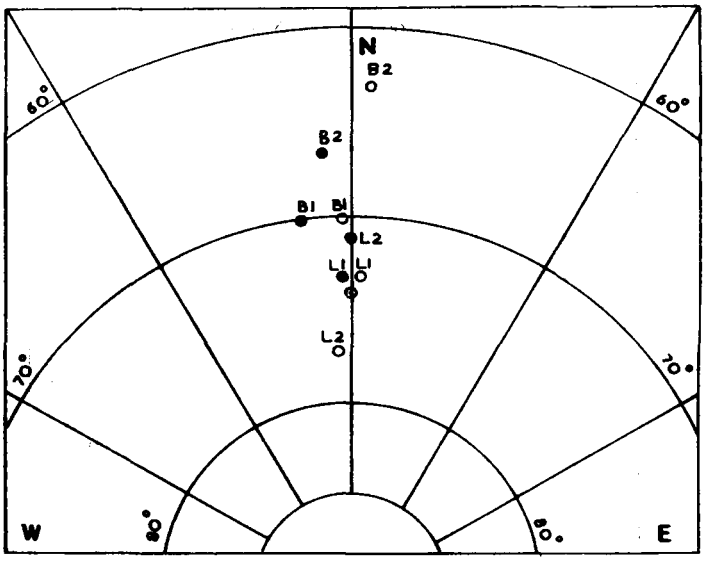


FIG. 6.—*Stereographic projection of polarization directions of varves from Ragunda.* ○ *uncorrected and* ● *corrected for dip. B1 and B2 indicate Bispfors Sections 1 and 2, whilst L1 and L2 indicate Lien Sections 1 and 2.*

Although the scatter in the Ragunda varves was comparable with that of the Prästmon series little of significance emerges from the measurements. These varves are about 9000 years old. Approximately the same hundred year period was sampled at all four sites, sets of six samples being taken from layers at twenty year intervals. The radii of the circles of confidence for the average polarization direction for the layers being comparable with the secular change over the hundred year interval, no information on this could be deduced from the data. Therefore, for comparison purposes, the mean polarization direction of all samples from each particular site has been calculated. These are shown in the stereographic projection in Fig. 6. The circles of confidence have radii

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of about 1°·5. The differences between the various values and the present field direction are just about large enough to suggest magnetic stability. There is an improvement in agreement when the means are calculated from the individually dip-corrected measurements, and with the exception of the average for section 2 at Bispfors the values are not inconsistent with each other. They may, therefore, be close to the mean direction of the Earth's field over the hundred year period of their deposition.

Some tentative conclusions may be arrived at as a result of the measurements described. On the basis of the Prästmon–Undrom and Ragunda data it is suggested that in a favourable environment the polarization directions acquired by varved clays on deposition will have directions which, on correction for the tilt of the layers, will lie close enough to the Earth's original field direction to give at least a rough approximation to the secular variation at the time. If the environment during deposition is unfavourable the polarization directions may be widely scattered and even systematically deviated, in which instance they may show little relation to the field direction. This appears to be the case with the Ångerman River delta sediments from which the cores were taken, and it seems probable that bottom currents are the factor most liable to create an unfavourable environment.

These conclusions are in the main substantiated by the laboratory sedimentation experiments of R. F. King (**8**) although there is one exception. Under natural conditions of sedimentation only the youngest sandy varves from the Ångerman River show the low inclinations to be expected from the experiments. This may be due to differences between sedimentation conditions in the laboratory and in nature. The low inclinations in the naturally deposited sandy varves may be due to the presence of a particularly high percentage of flat particles. Alternatively these low inclinations may be the result of the rolling of particles on the bottom by stream currents, and consequent rotation of the polarization direction in accord with the explanation given by R. F. King. The stream flow is certainly in the correct direction.

Differences between laboratory and natural conditions probably account also for discrepancies in magnitude between the amount of correction for geological dip derived from field measurements and from experiments.

Thus, assuming complete magnetic stability for the clays, the problem of obtaining reliable information on secular variation is one of finding out exactly what factors control the magnetization direction, and devising methods of making precise corrections for these effects. Hydrodynamic forces may well prove the most difficult to allow for, since correction for such forces will necessitate knowledge of the direction, and possibly the velocity, of the bottom currents during deposition. It seems possible, however, that by means of tank experiments a relation may be established between stream direction and velocity and the susceptibility anisotropy of the deposit. If this proves to be so, quantitative information about bottom currents during deposition may be obtained from susceptibility anisotropy measurements on the natural clays, and a previously determined correction made to allow for their effect on the polarization direction. The problem is of course complicated by the complexity of natural conditions, but it seems reasonable to expect that it may eventually prove possible by this means to secure approximate information about the periodicity and magnitude of the secular changes during the Pleistocene.

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